Lecture 7 Guided Self-Study

- Cache-Friendly C++
  - Jonathan Müller, CppCon 2025
  - Watch: 0:00–11:00 and 23:30–52:15 (~40 minutes)
- Reflect: use the study questions on the next slide.
- Focus: how cache lines, memory access patterns, data layout, and prefetching affect program performance.

Study Questions

1. In the opening example, why can a sorted std::vector outperform a hash table (std::unordered_set) despite having worse asymptotic lookup complexity?
2. Why can using smaller data types improve performance even when a program accesses each element only once?
3. Why is sequential array traversal generally more cache-friendly than pointer chasing? How does hardware prefetching contribute to the difference?
4. The talk compares Array of Structures (AoS) and Structure of Arrays (SoA). Why might SoA perform better when a program accesses only one field of each object? When might AoS be preferable?
5. What is false sharing? Why can two threads updating different variables interfere with each other, and how can data layout help avoid this problem?